

Predicting In Game Spending Behavior Using Machine Learning: An Economic and Behavioral
Analysis



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Author's note

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The Topic and Research Problem

Monetization has become a huge shift for the video game industry, and these transactions occur in many forms including: in game purchases, microtransactions, loot boxes, and virtual currencies. These mechanisms are now a huge part of industry revenues and especially for games that are free to play or games that are on mobile devices. While they have a huge economic importance, player spending behavior remains highly uneven. Only a small proportion of the total users account for a large share of the total revenue that a game makes. The majority of Player don't spend anything on games in terms of microtransactions and loot boxes therefore understanding the drivers of this behavior is a central issue for economists, data analysts, and most importantly game companies.

In terms of economic perspective, the microtransactions that occur in game are a representation of digital consumption, I like to call this idea of digital consumption under a big umbrella called the "The Digital Economy". I call it this because there is a whole new world of transactions, and finance in a way we did not imagine it before, and it is all done digitally with no physical interaction. Traditional Economic concepts such as utility maximization, price sensitivity, and consumer choices still occur here and studying how they interact with behavioral biases such as loss aversion, sunk costs and probabilistic rewards is extremely important and vital. I mention this because much of the existing literatures right now about this topic either focuses on theory or focus on virtual currencies like bitcoin and doge coin, while also having very minimal and limited economic interpretation.

In terms of the issue that is addressed in this research, I will address the lack of integrated studies that combine multiple things together like economic theory, behavioral analysis and data analytics all in one research to explain and predict in game spending behavior. At the player

level. I will be using machine learning techniques (explained later in the methodology) alongside economic reasoning, this study aims to bridge that gap and provide a clearer understanding of how player engagement and behavioral patterns translate into monetary decision in digital environments.

Research Question

The question that this research is built upon is:

Which player behaviors and engagement patterns best predict in game spending, and how can these predictors be interpreted using economic and behavioral theories of consumer choice?

Moving beyond simple descriptions and towards prediction and explanation is the reason why I chose this research question. I am not trying to answer whether players spend money or not, but rather I am focusing on behavioral factors that make spending more likely and interpreting these factors through established economic and analytical frameworks. The analysis will focus on many scopes which include:

- Player level behavioral data (e.g., playtime, activity frequency, progression),
- Spending outcomes related to microtransactions,
- A defined time window that allows prediction of future spending based on past behavior.

In this research I don't focus on game design or experimentation or psychological experiments, but rather I focus on observational data analysis. Maintaining a clear scope of the research is extremely important, The empirical analysis will concentrate on three main areas. First, the study will rely on player level behavioral data, including indicators such as playtime, activity frequency, progression metrics, and engagement intensity. These variables capture how players interact with the game over time and serve as proxies for commitment and perceived utility. Second, the analysis will focus on spending outcomes related to microtransactions, such

as total expenditure, purchase frequency, or a binary classification distinguishing spenders from non spenders. Third, the study will adopt a defined temporal framework, allowing past behavior to be used as input for predicting future spending, thereby avoiding purely contemporaneous or descriptive correlations.

Methodology

First the study will use quantitative data which is obtained in two ways:

- Public datasets from platforms like Kaggle, or public gameplay data
- I will personally request data sets from gaming companies like Ubisoft (This might not be feasible due to the nature of confidentiality gaming companies have)

These data sets will include a wide range of engagement indicators which include playtime, session frequency, activity intensity, progression, achievements, and retention related variables. In addition to this the analysis must be based on clear defined premises for spending behavior so measuring Monetary expenditure over a period of time, or including a binary classification to indicate whether people would spend money in games (1) or they would not spend money in games (0). These variables allow the study to capture both the depth and intensity of player engagement and link them directly to observed behavior in a structured and measurable way.

In addition to measuring and analyzing quantitative data, I also want to include a qualitative research component to complement and contextualize the empirical findings. Data in this case is even easier to find because online forums, reviews, and community discussions are all available online free and accessible. The problem will be authenticating that these reviews are coming from real people and gamers, and not bots that are implemented to deter information.

Adding this component to my research allows people to gain deeper insights into player's motivations, perception of value, and attitudes towards in game purchases and microtransactions, some things that might not be able to captured by numbers alone. These qualitative insights will be used to support and enrich the interpretation of the quantitative results, particularly in relation to behavioral economic mechanisms underlying spending decisions.

The analysis will proceed in several steps. First, data cleaning and preprocessing will be conducted to handle missing values, outliers, and inconsistencies, ensuring the reliability and validity of the dataset. Exploratory data analysis will then be used to examine distributions, correlations, and preliminary relationships between engagement variables and spending outcomes. Second, baseline econometric models, such as logistic regression, will be estimated to identify the key determinants of in game spending and to provide an interpretable benchmark grounded in economic theory. Third, machine learning models, including Random Forest and Gradient Boosting algorithms, will be applied to improve predictive performance and capture potential non linear relationships and interactions between variables. The results of these models will be evaluated using appropriate performance metrics and interpreted through economic and behavioral frameworks, with particular attention paid to feature importance, player heterogeneity, and differences in spending patterns across player segments.

Initial Literature Review

Initial readings reveal a growing and increasingly interdisciplinary body of literature at the intersection of gaming, economics, and data analytics. This literature reflects the rapid transformation of the video game industry toward monetization models centered on microtransactions and digital goods, raising important economic, behavioral, and ethical

questions. Several studies are dedicated to microtransaction and freemium monetization models, exploring their economic implications for both companies and consumers (Hamari et al., 2017; Zendle & Cairns, 2018). These studies highlight that it is not possible to fully describe the relationship between spending behaviour in digital games based on traditional relationships between price and game quality, as probabilistic reward structures and perceived value in games play a significant role in shaping the purchase decision (Hamari et al., 2017). In particular, loot boxes have been found to make significant use of randomness and reinforcement mechanisms, which can heighten the intensity of spending and raise ethical concerns because they are associated with gambling related behaviors (Zendle & Cairns, 2018).

Another strand of the literature takes a behavioral economic perspective on considering virtual consumption. Research on the sales of virtual items shows that purchase decisions are based on factors such as functional utility, social signaling and artificial scarcity, rather than pure rational cost benefit calculations (Lehdonvirta, 2009). Further work on the monetization of video games highlights the role of cognitive biases, such as sunk cost effects, loss aversion and reinforcement learning, in determining player spending behavior. (King & Delfabbro, 2018) Together, these studies lay a solid theoretical ground for understanding in game purchases as something that is often far from being rational choices (Lehdonvirta, 2009; King & Delfabbro, 2018).

A growing body of research also applies data analytics and machine learning techniques to the study of player behaviour. Predictive models have been proven to be successful in identifying behavioral outcomes such as churn and engagement based on player activity data (Perianez et al., 2016). While these approaches show good performance in terms of prediction,

they tend to focus more on the accuracy of the model than on economic interpretation, and hence questions about the relationship between the predictive variables and underlying consumer decision making processes remain open (Perianez et al., 2016).

This thesis contributes to these strands of literature by integrating machine learning based predictive modelling with economic and behavioural interpretation. Through combining quantitative prediction with theory based explanation, the paper strives to make a contribution towards a more holistic view of in game spending behavior as a measurable consequence and as an economically relevant decision (Hamari et al., 2017; Lehdonvirta, 2009; King & Delfabbro, 2018; Zendle & Cairns, 2018; Perianez et al., 2016).

Time Schedule

Period	Activity
Jan 2026	Finalize proposal and refine the research question
Mar- Apr 2026	Extended literature review
May – Jun 2026	Data collection and data cleaning
Jul – Aug 2026	Exploratory data analysis and baseline econometric models
Sep – Oct 2026	Machine learning modeling and model validation
Nov 2026	Results interpretation and discussion
Dec 2026	Writing, editing, and revisions
Jan 2027	Final submission

